

Serie 16

DETERMINANT

1. Show that the determinant of a 2×2 matrix is an alternating 2-linear function of rows and columns.
2. Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ be a 2×2 matrix over a field K .

(a) Show that the matrix A is invertible if and only if $ad - bc$ is not zero.

(b) Show that

$$A^2 = 0 \quad \text{if and only if} \quad \det(tI - A) = t^2$$

as polynomials in $K[t]$.

3. Each of the following expressions defines a function D on the set of 3×3 matrices over the field of real numbers. In which of these cases is D a 3-linear function?

(a) $D_1(A) = A_{11} + A_{22} + A_{33}$;

(b) $D_2(A) = A_{11}^2 + 3A_{11}A_{22}$;

(c) $D_3(A) = A_{11}A_{12}A_{33}$;

(d) $D_4(A) = A_{13}A_{22}A_{32} + 5A_{12}A_{22}A_{32}$;

(e) $D_5(A) = 0$;

(f) $D_6(A) = 1$.

4. Let K be a subfield of the complex numbers and n a positive integer. Let $j_1, \dots, j_n$ and $k_1, \dots, k_n$ be positive integers not exceeding n . For an $n \times n$ matrix A over K define

$$D(A) = A(j_1, k_1)A(j_2, k_2) \cdots A(j_n, k_n).$$

Prove that D is n -linear if and only if the integers $j_1, \dots, j_n$ are distinct.

5. Let K be a field. Define a function D on 3×3 matrices over K by the rule

$$D(A) = A_{11} \det \begin{bmatrix} A_{22} & A_{23} \\ A_{32} & A_{33} \end{bmatrix} - A_{12} \det \begin{bmatrix} A_{21} & A_{23} \\ A_{31} & A_{33} \end{bmatrix} + A_{13} \det \begin{bmatrix} A_{21} & A_{22} \\ A_{31} & A_{32} \end{bmatrix}.$$

Show that D is alternating and 3-linear as a function of the columns of A .